

Last time:

$$\left. \begin{array}{l} f: X \rightarrow Y \text{ continuous} \\ K \subseteq X \text{ compact} \end{array} \right\} \Rightarrow f(K) \subseteq Y \text{ compact}$$

• Take $Y = \mathbb{R}$:

Cor (Extreme value theorem) If $f: K \rightarrow \mathbb{R}$ is continuous and $K \neq \emptyset$ is a compact metric space, then f has a global minimum and a global maximum:
 $\exists a, b \in K$ s.t.

$$f(a) \leq f(x) \leq f(b) \quad \forall x \in K.$$

Pf: By the previous theorem, $f(K)$ is compact in $\mathbb{R}$. So $f(K)$ is closed and bounded. As $f(K)$ is bounded, $\inf f(K)$ and $\sup f(K)$ exist in $\mathbb{R}$.

Claim: $\sup f(K) \in \overline{f(K)}$. Suppose not: then $\exists \varepsilon > 0$ s.t. $(\sup f(K) - \varepsilon, \sup f(K) + \varepsilon) \cap f(K) = \emptyset$. But then $\sup f(K) - \frac{\varepsilon}{2}$ is a smaller upper bound for $f(K)$ — a contradiction.

Similarly, $\inf f(K) \in \overline{f(K)}$. Together:

$$\inf f(K), \sup f(K) \in \overline{f(K)} = f(K)$$

$\uparrow$ $f(K)$ closed

$$\Rightarrow \exists a, b \in K \text{ s.t. } f(a) = \inf f(K), \quad f(b) = \sup f(K).$$

□

Def Let X, Y be metric spaces and $f: X \rightarrow Y$. We

say f is uniformly continuous if: $\forall \varepsilon > 0, \exists \delta > 0$ s.t.

$$x, x_0 \in X, d_X(x, x_0) < \delta \Rightarrow d_Y(f(x), f(x_0)) < \varepsilon.$$

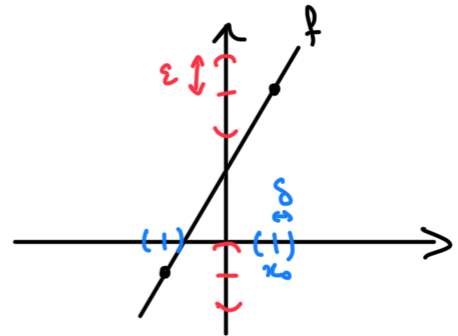
Rmk • Continuous $\Leftrightarrow \forall x_0 \in X, \forall \varepsilon > 0, \exists \delta > 0$ s.t. ...

$$\Leftrightarrow \delta = \delta(\varepsilon, x_0)$$

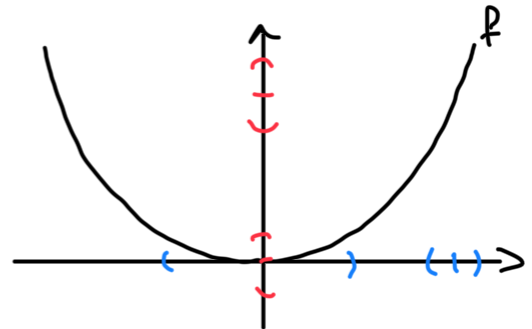
• Uniformly continuous $\Leftrightarrow \delta = \delta(\varepsilon)$.

• Uniformly continuous $\Rightarrow$ continuous
 $\nLeftarrow$

Ex ① $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 5x + 21$
is uniformly continuous



② $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$
is not uniformly continuous.



Prop If $f: X \rightarrow Y$ is continuous and X is compact, then f is uniformly continuous.

Pf: Fix $\varepsilon > 0$. As f is continuous, then for each $x_0 \in X \exists \delta_{x_0} > 0$ s.t.

$$d_X(x, x_0) < \delta_{x_0} \Rightarrow d_Y(f(x), f(x_0)) < \frac{\varepsilon}{2}.$$

Then $\{B_{\frac{1}{2}\delta_{x_0}}(x_0)\}_{x_0 \in X}$ is an open cover for X .

As X is compact, $\exists$ finitely many $x_1, \dots, x_n \in X$ s.t.

$$X \subseteq B_{\frac{1}{2}\delta_{x_1}}(x_1) \cup \dots \cup B_{\frac{1}{2}\delta_{x_n}}(x_n).$$

Set $\delta = \min \{ \frac{1}{2}\delta_{x_1}, \dots, \frac{1}{2}\delta_{x_n} \}$. Then $\delta > 0$, and

$$x, x_0 \in X, \quad d_X(x, x_0) < \delta \leq \frac{1}{2}\delta_{x_i}$$

$$\Rightarrow x_0 \in B_{\frac{1}{2}\delta_{x_i}}(x_i)$$

for some $1 \leq i \leq n$

$$\Rightarrow d_X(x, x_i) \leq d_X(x, x_0) + d_X(x_0, x_i)$$

$$< \frac{1}{2}\delta_{x_i} + \frac{1}{2}\delta_{x_i}$$

$$= \delta_{x_i}$$

$$\Rightarrow d_X(x, x_i) < \delta_{x_i} \quad \text{and} \quad d_X(x_i, x_0) < \frac{1}{2}\delta_{x_i} < \delta_{x_i}$$

$$\Rightarrow d_Y(f(x), f(x_i)) < \frac{\varepsilon}{2} \quad \text{and} \quad d_Y(f(x_i), f(x_0)) < \frac{\varepsilon}{2}$$

$$\begin{aligned} \Rightarrow d_Y(f(x), f(x_0)) &\leq d_Y(f(x), f(x_i)) + d_Y(f(x_i), f(x_0)) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

So f is uniformly continuous. $\square$

SEQUENCES OF FUNCTIONS (Rudin Ch. 7, Copson Ch. 7)

Let (X, d_X) and (Y, d_Y) be metric spaces, and suppose that $f_n: X \rightarrow Y$ is a function $\forall n \in \mathbb{N}$.

Def We say that the sequence $\{f_n\}_{n \geq 1}$ of functions converges pointwise to $f: X \rightarrow Y$ if: $\forall x \in X$, the sequence $\{f_n(x)\}_{n \geq 1}$ in Y converges to $f(x)$. In other words: $\forall x \in X, \forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t.

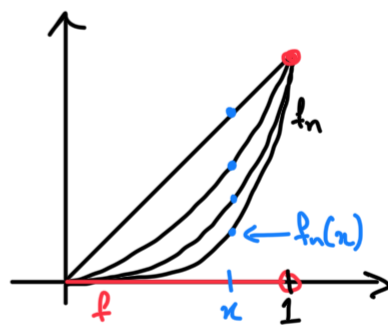
$$d_Y(f_n(x), f(x)) < \varepsilon \quad \forall n \geq N.$$

Ex $f_n: [0,1] \rightarrow \mathbb{R}$, $f(x) = x^n$ for $n \geq 1$.

Claim: f_n converges pointwise

to

$$f(x) = \begin{cases} 0 & \text{if } x \in [0,1) \\ 1 & \text{if } x = 1 \end{cases}$$



- Case: $x \in [0,1)$. Then $f_n(x) = x^n \rightarrow 0$ as $n \rightarrow \infty$, since $x < 1$.

- Case: $x = 1$. Then $f_n(x) = 1^n = 1 \rightarrow 1$.

Rmk So:

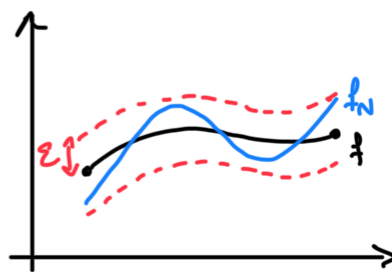
$$\left. \begin{array}{l} f_n \text{ continuous} \\ f_n \text{ converges pointwise to } f \end{array} \right\} \not\Rightarrow f \text{ continuous}$$

- We need something stronger here...

Def We say that the sequence of functions $\{f_n\}_{n \geq 1}$ converges uniformly to $f: X \rightarrow Y$ if:

$\forall \varepsilon > 0 \exists N \in \mathbb{N}$ s.t.

$$d_Y(f_n(x), f(x)) < \varepsilon \quad \forall n \geq N \quad \text{and} \quad \forall x \in X.$$



Rmk • Converges pointwise $\Leftrightarrow \forall x \in X, \forall \varepsilon > 0, \exists N \dots$
 $\Leftrightarrow N = N(\varepsilon, x)$

- Converges uniformly $\Leftrightarrow N = N(\varepsilon)$

• Converges uniformly $\Rightarrow$ converges pointwise
 $\nLeftarrow$